Math 520, F07 **Name:** Answer
Quiz 4 **SSN:** xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. You must simplify your answer if possible.

Problem 1. (10 points) Construct the Picard iterates for the initial-value problem

$$y' = -y, \quad y(0) = 1,$$

and find the limit of these iterates.

Solution: $f(t, y) = -y$.

$$y(t) = y(0) + \int_{t_0}^{t} f(s, y(s))ds$$
$$= 1 + \int_0^t (-y(s))ds$$

Picard iterates:

$$y_1(t) = 1 + \int_0^t -y_0(s)\, ds = 1 - \int_0^t ds = 1 - t$$
$$y_2(t) = 1 + \int_0^t -y_1(s)\, ds = 1 + \int_0^t (-1+s) ds = 1 - t + \frac{t^2}{2}$$
$$y_3(t) = 1 + \int_0^t -y_2(s) ds = 1 + \int_0^t \left(-1 + s - \frac{s^2}{2}\right) ds$$
$$= 1 - t + \frac{t^2}{2} - \frac{t^3}{3!}$$

$\cdots$

$$y_n(t) = 1 + \int_0^t -y_{n-1}(s) ds = 1 + \int_0^t -\left[1 - s + \frac{s^2}{2} - \frac{s^3}{3!} + \cdots + \frac{(-s)^{n-1}}{(n-1)!}\right] ds$$
$$= 1 - t + \frac{t^2}{2} - \frac{t^3}{3!} + \cdots + \frac{(-t)^n}{n!}$$

$$\lim_{n\to\infty} y_n(t) = \lim_{n\to\infty} \left[1 - t + \frac{t^2}{2} - \frac{t^3}{3!} + \cdots + \frac{(-t)^n}{n!}\right]$$
$$= e^{-t}$$

Problem 2. (10 points) Show that the solution $y(t)$ of the following initial-value problem

$$y' = 1 + y + y^2 \cos t, \quad y(0) = 0,$$

exists on the interval $0 \le t \le \frac{1}{3}$.

Solution: Here $t_0 = 0$, $y_0 = 0$, $a = \frac{1}{3}$.

Let R be the rectangle $0 \le t \le \frac{1}{3}$, $|y| \le b$ for some $b > 0$.

$$M = \max_{(t,y)\in R} |f(t,y)| = \max_{\substack{0 \le t \le \frac{1}{3} \\ |y| \le b}} |1 + y + y^2 \cos t|$$

$$= 1 + b + b^2$$

$$\alpha = \min\left(a, \frac{b}{M}\right) = \min\left(\frac{1}{3}, \frac{b}{1+b+b^2}\right)$$

Notice that

$$\frac{b}{1+b+b^2} = \frac{1}{\frac{1}{b} + 1 + b} = \frac{1}{(\frac{1}{b} + b) + 1} \le \frac{1}{3}$$

where the equality holds when $b = 1$ (i.e., $\frac{1}{b} = b$)

Let us choose $b = 1$, then

$$\alpha = \min\left(\frac{1}{3}, \frac{1}{3}\right) = \frac{1}{3}$$

Thus, $y(t)$ exists on the interval $0 \le t \le \frac{1}{3}$.